

115 年度 社團法人中華產業機械設備協會 推廣費曼學習記錄

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日期 115 年 6 月 9 日

1. 學習項目：ASME 之 PV-ELITE 強度計算書檢查重點及稽核實務概述
2. 學習動機：為了解 PV-ELITE 強度計算書計算內容，以利了解設備設計資訊，方便與設計圖面核對，確保設備製造與計算書相符。
3. 績效與效益：
 1. 了解各章節計算內容及閱讀重點。
 2. 學習焊道分級、類型，與相對應的焊接效率數值。
 3. 了解同樣材料間的應力差異的關係。

學習相片剪影



ASME之PV-ELITE強度計算書

檢查重點及稽核實務概述

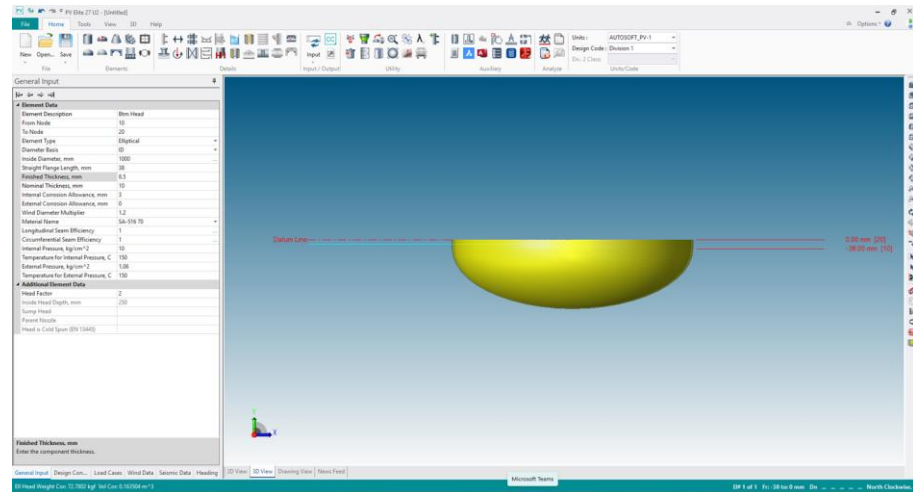
報告人:王詠民
115年6月9日



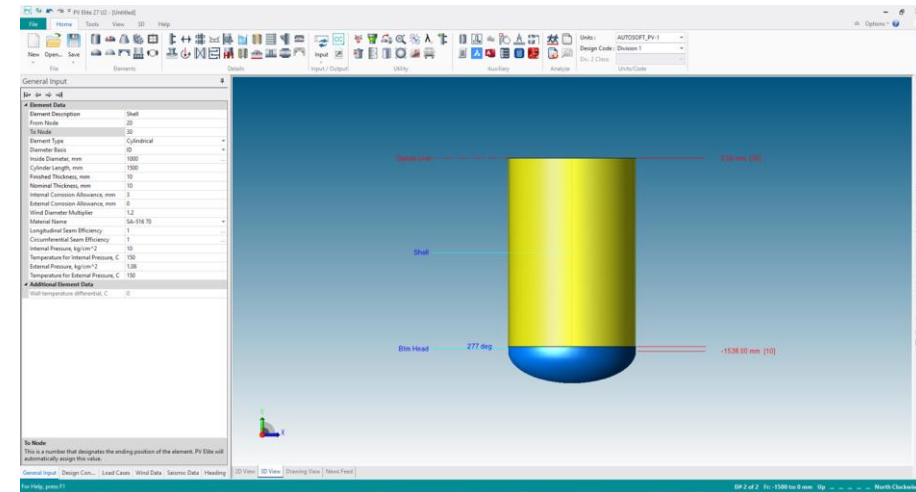
PV-ELITE 軟體建模順序

常見順序:由下至上，由左至右

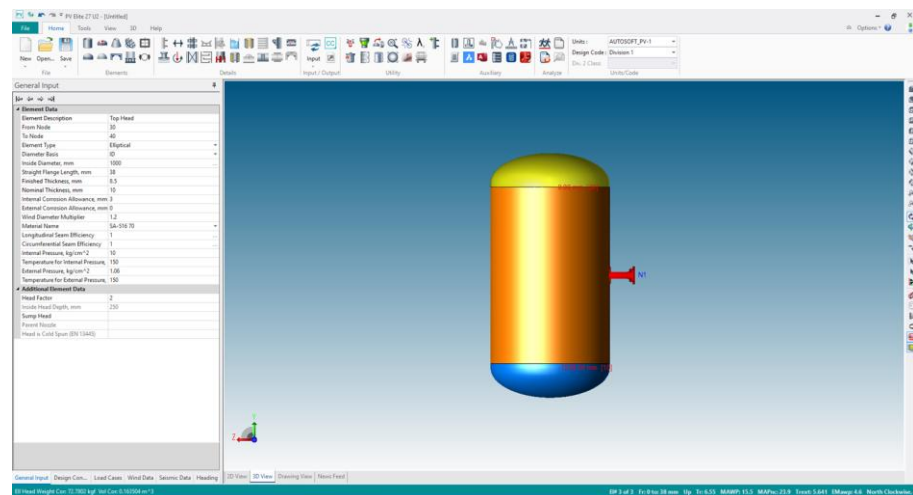
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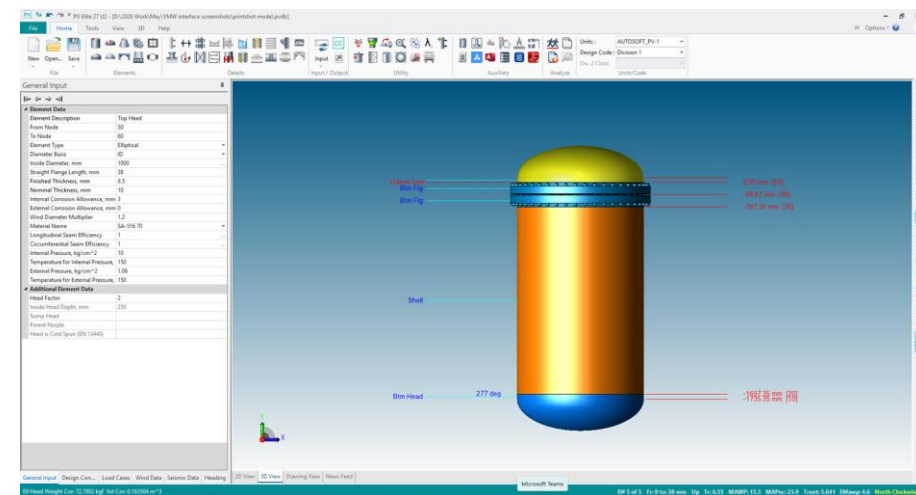
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4.



PV-ELITE 強度計算書目錄檢核

第一部分：主體設計

Vessel Design Summary / 設計條件一覽
Nozzle Summary / 插管一覽
Nozzle Schedule / 插管規格
Input Echo / 基本條件輸入
Flg Calc [Int P] / 胴體法蘭詳細數據計算
Internal Pressure Calculations / 承受內壓強度計算
Element / 主體部件計算
Shop Pressure Test Results / 耐壓試驗數值
External Pressure Calculations / 承受外壓強度計算

第二部分：載重、管嘴與附加

Nozzle Flange MAWP / 管嘴法蘭之MAWP限界
Wind & Earthquake Calc / 風載與耐震強度算式
Saddle Calcs (Operating/Test) / 操作/試壓狀況下鞍座應力
Nozzle Calcs / 插管計算
ASME TS Calc / 管板強度計算

設計條件一覽 (Vessel Design Summary)

ASME Code, Section VIII Division 1, 2025

Diameter Spec : 219.100 mm OD
Vessel Design Length, Tangent to Tangent 2133.80 mm

Specified Datum Line Distance 0.00 mm

Shell Side Design Temperature	40	C
Channel Side Design Temperature	100	C
Shell Side Design Pressure	6.000	kg/cm2
Channel Side Design Pressure	8.800	kg/cm2
Shop Shell Side Test Pressure	7.800	kg/cm2
Shop Channel Side Test Pressure	11.523	kg/cm2

Wind Design Code ASCE/SEI 7-22
Earthquake Design Code G-Loading

Shell : 胴側
Channel : 管側

ASME 材質核對與 UNS 命名規則

Materials of Construction:

Component Type	Material	Class	Thickness	UNS #
Shell	SA-403 304	...	...	S30400
Shell	SA-312 TP304	...	...	S30400
Head	SA-403 304	...	...	S30400
Flange	SA-182 F304	...	<= 130	S30400
Nozzle	SA-312 TP304	...	...	S30400
Nozzle	SA-182 F304	...	<= 130	S30400
Nozzle Flg	SA-182 F304	...	> 130	S30400
Nozzle Flg	SA-182 F304	...	<= 130	S30400
Tubes	SA-213 TP304	...	...	S30400
Tubesheet	SA-182 F304	...	<= 130	S30400
Flg Bolting	SA-193 B7	...	<= 64	G41400
Hrz Bolting	SA-193 B7	...	<= 64	G41400
Saddles	SA-36	...	...	K02600

UNS (統一編號系統) 首字母分類：



S 字頭： 代表不鏽鋼及耐熱鋼 (Stainless Steels)



N 字頭： 代表鎳及鎳合金 (Nickel and Alloys)



C 字頭： 代表銅及銅合金 (Copper and Alloys)



A 字頭： 代表鋁及鋁合金 (Aluminum and Alloys)

主體部件厚度設計厚度

Element Type	"To" Elev mm	Element Length mm	Nominal Thickness mm	Finished Thickness mm	Reqd Thk Internal mm	Reqd Thk External mm	Long Eff	Circ Eff
Ellipse	0.0	0.0	3.8	3.3	1.5	0.6	1.00	1.00
Cylinder	47.2	47.2	3.8	3.3	1.5	0.4	1.00	1.00
Body Flg	150.4	100.0	...	28.6	19.3	19.3	1.00	1.00
Cylinder	1955.4	1777.0	3.8	3.3	1.5	1.3	0.85	0.85
Body Flg	2086.6	100.0	...	28.6	19.3	19.3	1.00	1.00
Cylinder	2133.8	47.2	3.8	3.3	1.5	0.4	1.00	1.00
Ellipse	2133.8	0.0	3.8	3.3	1.5	0.6	1.00	1.00

* 注意：此處厚度數值已四捨五入

鞍座參數 (Saddle Parameters)

Saddle Parameters:

Saddle Width	75.000	mm
Saddle Bearing Angle	120.000	deg.
Centerline Dimension	300.000	mm
<u>Wear Pad</u> Width	120.000	mm
Wear Pad Thickness	5.000	mm
Wear Pad Bearing Angle	152.000	deg.
Distance from Saddle to Tangent	285.500	mm

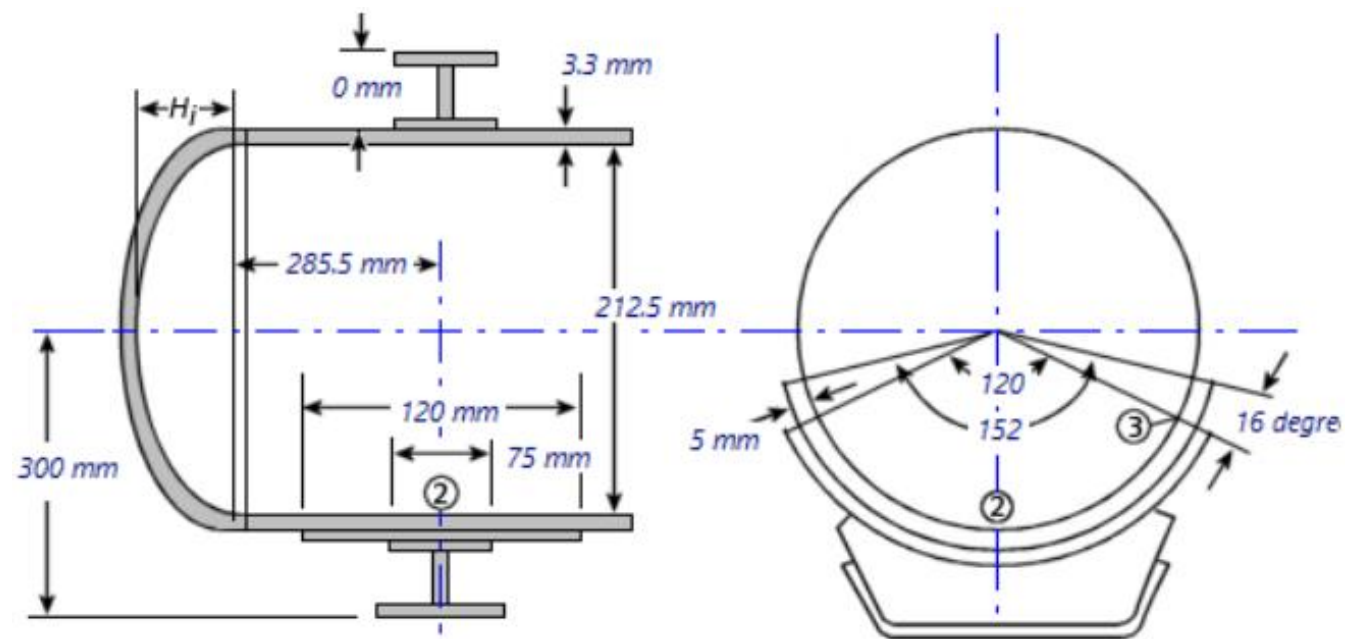
Wear Pad：補強板

Baseplate：底板

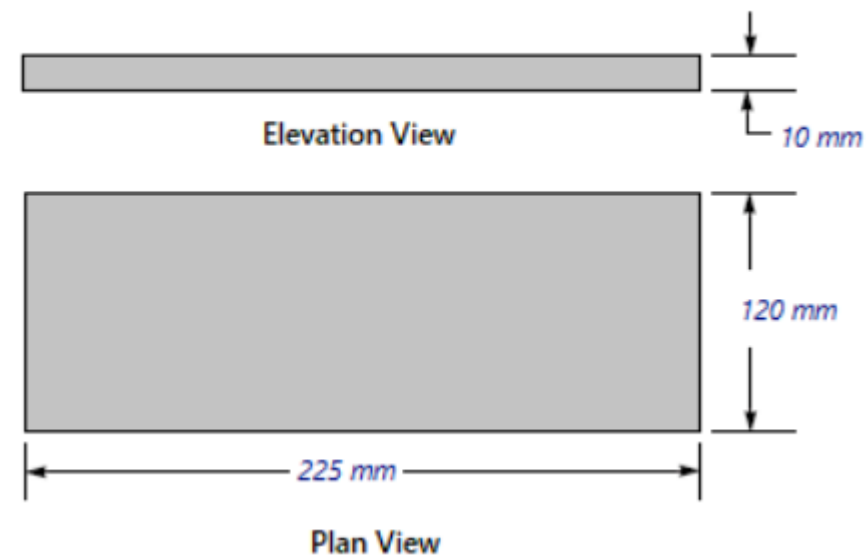
Rib：肋板

<u>Baseplate</u> Length	225.000	mm
Baseplate Thickness	10.000	mm
Baseplate Width	120.000	mm
Number of Ribs (including outside ribs)	2	
<u>Rib</u> Thickness	6.000	mm
Web Thickness	6.000	mm
Height of Center Web	175.000	mm
Number of Bolts in Baseplate	2	

鞍座圖示



Baseplate Sketch



插管一覽(Nozzle Summary)

Check the spatial relationship between the nozzles:

From Node	Nozzle Description	X Coordinate mm	Layout Angle deg	Dia. Limit mm
20	N1	23.600	0.000	47.000
20	N2	23.600	180.000	47.000
40	N4	247.400	0.000	48.616
40	N3	1880.400	180.000	48.616
40	D2	247.400	180.000	52.836
40	V2	1880.400	0.000	52.836
40	S	442.400	0.000	52.836
60	V1	2110.200	0.000	47.000
60	D1	2110.200	180.000	47.000

插管規格(Nozzle Schedule)

Nozzle Schedule:

Description	Nominal or Actual Size	Schd or FVC Type	Flg Type	Nozzle O/Dia in	Wall Thk mm	Reinforcing Diameter	Pad Thk mm	Cut Length mm	Flg Class
N1	1.000 in	80S	SlipO	1.315	4.547	...	...	165.36	150
N2	1.000 in	80S	SlipO	1.315	4.547	...	...	165.36	150
N4	1.000 in	80S	SlipO	1.315	4.547	...	...	101.22	150
N3	1.000 in	80S	SlipO	1.315	4.547	...	...	101.22	150
V1	1.375 in	Actual	None	1.375	4.128	...	...	52.25	...
D1	1.375 in	Actual	None	1.375	4.128	...	...	52.25	...
D2	1.377 in	Actual	None	1.377	4.279	...	...	55.75	...
V2	1.377 in	Actual	None	1.377	4.279	...	...	52.26	...
S	1.377 in	Actual	None	1.377	4.279	...	...	52.26	...

插管規格(Nozzle Schedule)

Description	Material	Shl Grve Weld	Noz Shl/Pad Weld	Pad OD Weld	Pad Grve Weld	Inside Weld
N1	SA-312 TP304	3.290	4.000	...	...	...
N2	SA-312 TP304	3.290	4.000	...	...	...
N4	SA-312 TP304	3.290	4.000	...	...	...
N3	SA-312 TP304	3.290	4.000	...	...	...
V1	SA-182 F304	3.290	4.000	...	...	...
D1	SA-182 F304	3.290	4.000	...	...	...
D2	SA-182 F304	3.290	5.000	...	...	...
V2	SA-182 F304	3.290	5.000	...	...	...
S	SA-182 F304	3.290	5.000	...	...	...

輸入内容(Input Echo)

PV Elite Vessel Analysis Program: Input Data

Exchanger Design Pressures and Temperatures

Shell Side Design Pressure	6	kg/cm2
Channel Side Design Pressure	8.8002	kg/cm2
Shell Side Design Temperature	40.0	C
Channel Side Design Temperature	100.0	C
Radiography, Shell Side	RT-3	
Radiography, Channel Side	RT-1	
Service Type, Shell Side	None	
Service Type, Channel Side	None	
MDMT (CET), Shell Side	-19.0	C
MDMT (CET), Tube Side	-19.0	C
User defined MAWP, Shell Side	6	kg/cm2
User defined MAWP, Channel Side	8.8002	kg/cm2
User defined MAPnc, Shell Side	0	kg/cm2
User defined MAPnc, Channel Side	0	kg/cm2
User defined Test Pres., Shell Side	0	kg/cm2
User defined Test Pres., Channel Side	0	kg/cm2

銲接級別 (RT1~RT4) 與效率查核

 **RT 1**：所有 縱向(A類)、周向(B、C 類)對接焊道進行全射線檢測(Full Radiography)。

焊道效率 $E=1.0$ 。

常用於高壓或致死物之高安全等級容器。

 **RT 2**：所有縱向(A 類)對接焊道全射線檢測(Full Radiography)。

周向(B、C類)對接焊道進行局部射線檢查(Spot Radiography)。

焊道效率 $E=1.0$ 。

 **RT 3**：縱向及周向焊道(A、B類)都只進行局部射線檢查(Spot Radiography)。

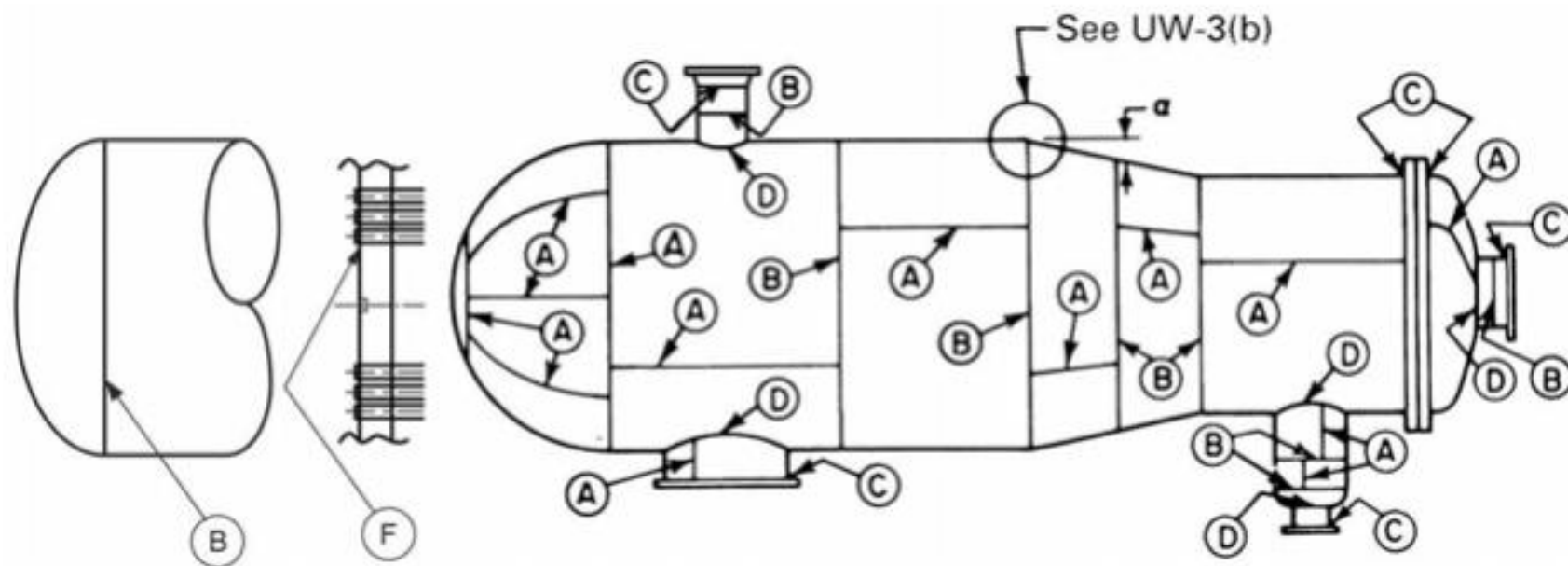
焊道效率 $E=0.85$ 。

 **RT 4**：不完全符合上述RT 1~RT3要求時就歸類在這邊。

焊道效率 $E=0.7$ or 0.85 。

銲道分類圖示

Figure UW-3
Illustration of Welded Joint Locations Typical of Categories A, B, C, D, and F



基本條件輸入-Shell(1/2)

Element#	4/7	
Element From Node	40	
Element To Node	50	
Element Type	Cylinder	
Description	Shell	
Distance "FROM" to "TO"	1777	mm
Element Outside Diameter	219.1	mm
Element Thickness	3.29	mm
Internal Corrosion Allowance	0	mm
Nominal Thickness	3.76	mm
External Corrosion Allowance	0	mm
Design Internal Pressure	6	kg/cm ²
Design Temperature Internal Pressure	40	C
Design External Pressure	0.53	kg/cm ²

Element#：元件節點:端板、胴體、胴體法蘭、蓋板、圓錐體。

Element From/To Node：元件起始/終止節點

Nominal Thickness：實際材料使用厚度

Element Thickness：扣除公差後的厚度

基本條件輸入-Shell(2/2)

Design Temperature	External Pressure	40	C
Effective Diameter Multiplier		1.2	
Material Name		SA-312	TP304
Allowable Stress, Ambient		1192.9	Kg/cm2
Allowable Stress, Operating		1192.9	Kg/cm2
Allowable Stress, Hydrotest		1881.1	Kg/cm2
Material Density		8027.2	kg/m3
P Number Thickness		0	mm
Yield Stress, Operating		2110.5	Kg/cm2
External Pressure Chart Name		HA-1	
UNS Number		S30400	
Product Form		Wld. pipe	
Efficiency, Longitudinal Seam		0.85	
Efficiency, Circumferential Seam		0.85	
Weld is pre-Heated		No	

有縫管/無縫管材料應力

Line No.	Nominal Composition	Product Form	Spec. No.	Type/Grade	Alloy Desig./ UNS No.	Class/ Condition/ Temper	Size/ Thickness, mm	P-No.	Group No.
31	18Cr-8Ni	Smls. & wld. pipe	SA-312	TP304	S30400	...	...	8	1
32	18Cr-8Ni	Wld. pipe	SA-312	TP304	S30400	...	...	8	1
33	18Cr-8Ni	Wld. pipe	SA-312	TP304	S30400	...	...	8	1

Line No.	Min. Tensile Strength, MPa	Min. Yield Strength, MPa	Applicability and Max. Temperature Limits (NP = Not Permitted) (SPT = Supports Only)				External Pressure Chart No.	Notes
			I	III	VIII-1	XII		
31	515	205	816	NP	816	343	HA-1	G12, T8, W13, W14
32	515	205	816	NP	816	343	HA-1	G3, G5, G12, G24, T7
33	515	205	816	NP	816	343	HA-1	G3, G12, G24, T8

Line No.	Maximum Allowable Stress, MPa (Multiply by 1000 to Obtain kPa), for Metal Temperature, °C, Not Exceeding														
	40	65	100	125	150	200	250	300	325	350	375	400	425	450	475
31	138	126	113	107	103	95.7	89.9	85.9	84.1	82.2	80.5	79.2	77.3	76.0	74.8
32	117	117	117	114	111	107	103	98.5	96.5	94.7	92.8	90.9	89.1	87.1	85.9
33	117	107	96.2	91.2	87.4	81.1	76.6	72.7	71.2	70.0	68.8	67.5	66.3	65.0	63.8

G3 These stress values include a joint efficiency factor of 0.85.

G5 Due to the relatively low yield strength of these materials, these higher stress values were established at temperatures where the short-time tensile properties govern to permit the use of these alloys where slightly greater deformation is acceptable. The stress values in this range exceed $66\frac{2}{3}\%$ but do not exceed 90% of the yield strength at temperature. Use of these stresses may result in dimensional changes due to permanent strain. These stress values are not recommended for the flanges of gasketed joints or other applications where slight amounts of distortion can cause leakage or malfunction. For Section III applications, Table Y-2 lists multiplying factors that, when applied to the yield strength values shown in Table Y-1, will give allowable stress values that will result in lower levels of permanent strain.

主體上的配件-鞍座

Element From Node	40	
Detail Type	Saddle	
Detail ID	Left Saddle	
Dist. from "FROM" Node / Offset dist	285.5	mm
Width of Saddle	75	mm
Height of Saddle at Bottom	300	mm
Saddle Contact Angle	120.0	
Height of Composite Ring Stiffener	0	mm
Width of Wear Plate	120	mm
Thickness of Wear Plate	5	mm
Contact Angle, Wear Plate (degrees)	152.0	
Friction coefficient	0.0	
Moment Factor	3.0	
Dimension E at base (optional)	0	mm
Circumferential Eff. over Saddle	1.0	
Circumferential Eff. at Midspan	1.0	
Tangent to Tangent dist. (optional)	0	mm

主體上的配件-插管

Element From Node	40	
Detail Type	Nozzle	
Detail ID	D2	
Dist. from "FROM" Node / Offset dist	69	mm
Nozzle Diameter	1.377	in.
Nozzle Schedule	None	
Nozzle Class	None	
Layout Angle	180.0	
Blind Flange (Y/N)	N	
Weight of Nozzle (Used if > 0)	0	kgf
Grade of Attached Flange	None	
Nozzle Matl	SA-182 F304	

胴體法蘭詳細數據計算(Flg Calc [Int P])-主體尺寸

Lft Flg

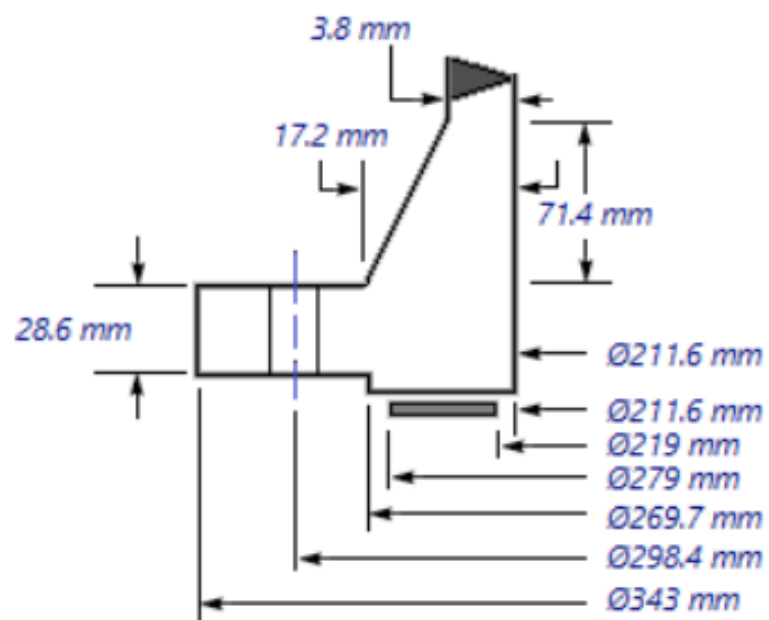
Flange Type	Integral Weld Neck		
Design Pressure	P	8.82	kg/cm ²
Design Temperature		100	C
Internal Corrosion Allowance	ci	0.0000	mm
External Corrosion Allowance	ce	0.0000	mm
Use Corrosion Allowance in Thickness Calcs.		Yes	
Flange Inside Diameter	B	211.580	mm
Flange Outside Diameter	A	343.000	mm
Flange Thickness	t	28.6000	mm
Thickness of Hub at Small End	go	3.7600	mm
Thickness of Hub at Large End	g1	17.2100	mm
Length of Hub	h	71.4000	mm

胴體法蘭詳細數據計算(Flg Calc [Int P])-法蘭材質、螺栓資料

Flange Material		SA-182 F304	
Flange Material UNS number		S30400	
Flange Allowable Stress At Temperature	Sfo	1152.14	Kg/cm2
Flange Allowable Stress At Ambient	Sfa	1406.99	Kg/cm2
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1753.64	Kg/cm2
Bolt Allowable Stress At Ambient	Sa	1753.64	Kg/cm2
Diameter of Bolt Circle	C	298.400	mm
Nominal Bolt Diameter	a	19.0500	mm
Type of Threads	TEMA Thread Series		
Number of Bolts		12	

胴體法蘭詳細數據計算(Flg Calc [Int P])-墊片資料

Flange Face Outside Diameter	Fod	269.700	mm
Flange Face Inside Diameter	Fid	211.580	mm
Flange Facing Sketch		Sketch 1a	
Gasket Outside Diameter	Go	279.000	mm
Gasket Inside Diameter	Gi	219.000	mm
Gasket Factor	m	2.0000	
Gasket Design Seating Stress	y	112.48	Kg/cm2
Gasket Thickness	tg	3.2000	mm
Raised face depth	Fd	0.0000	mm
Length of Partition Gasket	lp	409.0000	mm
Width of Partition Gasket	tp	6.0000	mm
Partition Gasket Factor	mPart	2.0000	
Partition Gasket Design Seating Stress	yPart	112.48	Kg/cm2



承受內壓強度計算(Internal Pressure Calculations:)

Element Thickness, Pressure, Diameter and Allowable Stress :

From	To	Int. Press + Liq. Hd kg/cm ²	Nominal Thickness mm	Total Corr Allowance mm	Element Diameter mm	Allowable Stress(SE) Kg/cm ²	Element#
	Lft Cap	8.8207	3.76	...	219.1	1396.8	1
	Cap SF	8.8	3.76	...	219.1	1396.8	2
	Lft Flg	8.8203	...	...	211.58	1152.1	3
	Shell	6.0212	3.76	...	219.1	1014	4
	Rgh Flg	8.8205	...	...	211.58	1152.1	5
	Rgh Cap SF	8.8	3.76	...	219.1	1396.8	6
	Rgh Cap	8.8207	3.76	...	219.1	1396.8	7

Element Required Thickness and MAWP :

From	To	Design Pressure kg/cm ²	M.A.W.P. Corroded kg/cm ²	M.A.P. New & Cold kg/cm ²	Minimum Thickness mm	Required Thickness mm	Element#
	Lft Cap	8.8	43.1	43.43	3.29	1.5	1
	Cap SF	8.8	42.46	42.77	3.29	1.5	2
	Lft Flg	8.8	39.85	48.69	28.6	19.3294	3
	Shell	6	30.8	30.82	3.29	1.5	4
	Rgh Flg	8.8	40.08	48.97	28.6	19.2532	5
	Rgh Cap SF	8.8	42.46	42.77	3.29	1.5	6
	Rgh Cap	8.8	43.1	43.43	3.29	1.5	7

主體部件計算(Element: Shell)

Internal Pressure Results for: Shell

Cylindrical Shell Design Information:				
Design Pressure	P	6.021 kg/cm^2	Int. Design Temperature	40 C
Material		SA-312 TP304	Ext. Design Temperature	40 C
Length		<u>1777.000 mm</u>	External Pressure Chart	HA-1
			UNS Number	S30400
Int. Corr. All.	c	0.0000 mm	Allowable Stress (ope)	S 1192.9 Kg/cm2
Ext. Corr. All.	ce	0.0000 mm	Allowable Stress (amb)	Sa 1192.9 Kg/cm2
Outside Diameter	Do	219.100 mm	Actual Stress	233.0 Kg/cm2
Spec. Min. Thk t,e, or sa		<u>3.2900 mm</u>	Spec. Nominal Thk.	<u>3.760 mm</u>
Surface Area		1228397.6 mm2	SG of Contents	1.000
Element Volume		0.0 m^3	Weight of Contents	44.8 kgf
Empty Weight		36.4 kgf	Operating Weight	81.2 kgf
			Comp. All. Ope. Stress	1077.1 Kg/cm2

主體部件計算(Element: Shell)

Radiography Information:				
User Defined		User Defined		
Circ. Joint Efficiency	85 %	Long. Joint Eff.	E or v	85 %

Required Thickness due to Internal Pressure [tr]:

= $(P \cdot Ro) / (S \cdot E + 0.4 \cdot P)$ per Appendix 1-1 (a)(1)

= $(6.021 \cdot 109.5) / (1193 \cdot 0.85 + 0.4 \cdot 6.021)$

= $0.6490 + 0.0000 = 0.6490 \text{ mm}$

Note: The thickness required was less than the Code Minimum, therefore the Code Minimum value of 1.5000 mm per UG-16 will be used.

耐壓試驗數值(Shop Pressure Test Results)

Hydrostatic Test Pressure Results:

Exchanger Shell Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * <u>Sa/S</u>	7.800 kg/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	7.800 kg/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	25.320 kg/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	6.600 kg/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	8.580 kg/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	8.580 kg/cm ²
App 27-4	= M.A.W.P.	6.000 kg/cm ²

UG99(b):規定水壓試驗最小值為最高使用壓力1.3倍+溫度修正

(35):可將MAWP視為與設計壓力相同

此案M.A.P. 受到插管法蘭限制

Exchanger Channel Side Hydrostatic Test Pressures:

UG99(b)	= 1.30 * M.A.W.P. * Sa/S	11.523 kg/cm ²
UG99b(b)(35)	= 1.30 * Design Pres * Sa/S	11.523 kg/cm ²
UG99(c)	= 1.30 * M.A.P. - Head(Hyd)	25.320 kg/cm ²
UG100	= 1.10 * M.A.W.P. * Sa/S	9.751 kg/cm ²
PED	= max(1.43*DP, 1.25*DP*ratio)	12.584 kg/cm ²
PED&UG99(b)(35)	= max(1.43*DP, 1.3*DP*ratio)	12.584 kg/cm ²
App 27-4	= M.A.W.P.	8.800 kg/cm ²

35 The maximum allowable working pressure may be assumed to be the same as the design pressure when calculations are not made to determine the maximum allowable working pressure.

耐壓試驗數值(Shop Pressure Test Results)

UG-99(b)(35), Test Pressure Calculation [Shell Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 6 • 1

= 7.800 kg/cm²

UG-99(b) Note 35, Test Pressure Calculation [Channel Side]:

= Test Factor • Design Pressure • Stress Ratio

= 1.3 • 8.8 • 1.007

= 11.523 kg/cm²

試壓狀況下鞍座應力(Saddle Calcs: Test Case)

Input and Calculated Values:

Vessel Mean Radius	Rm	107.90	mm
Shell Thickness used in this Case	t	3.290	mm
Stiffened Vessel Length per 4.15.6	L	1777.00	mm
Distance from Saddle to Vessel tangent	a1 or a	285.50	mm
Saddle Width	b1 or b	75.00	mm
Saddle Bearing Angle	delta or theta	120.00	degrees
Wear Plate Width	b2 or b1	120.00	mm
Wear Plate Bearing Angle	delta2 or theta1	152.00	degrees
Wear Plate Thickness	e2 or tr	5.0	mm
Wear Plate Allowable Stress	fw or Sr	1192.89	Kg/cm2
Shell Allowable Stress used in Calculation		1881.09	Kg/cm2
Head Allowable Stress used in Calculation		1881.09	Kg/cm2
Circumferential Efficiency in Plane of Saddle		1.00	
Circumferential Efficiency at Mid-Span		1.00	
Distance from Saddle Base to Centerline	B	300.00	mm
Coefficient of Friction	mu	0.00	

鞍座底板及肋板尺寸

Results for Vessel Ribs, Web and Base:

Baseplate Length	Bplen	225.0000	mm
Baseplate Thickness	Bpthk	10.0000	mm
Baseplate Width	Bpwid	120.0000	mm
Number of Ribs (inc. outside ribs)	Nribs	2	
Rib Thickness	Ribtk	6.0000	mm
Web Thickness	Webtk	6.0000	mm
Web Location	Webloc	Side	
Saddle Yield Stress	Sy	2548.9	Kg/cm2
Height of Web at Center	Hw,c	175.0	mm
Friction Coefficient	mu	0.000	

鞍座底板螺栓計算

Input Data for Base Plate Bolting Calculations:

Total Number of Bolts per BasePlate	Nbolts	2
Total Number of Bolts in Tension/Baseplate	Nbt	1
Bolt Material Specification		SA-193 B7
Bolt Allowable Stress	Stba	1753.64 Kg/cm2
Bolt Corrosion Allowance	Bca	0.0 mm
Distance from Bolts to Edge	Edgedis	52.5 mm
Nominal Bolt Diameter	Bnd	19.0500 mm
Thread Series	Series	TEMA
BasePlate Allowable Stress	S	1162.30 Kg/cm2
Area Available in a Single Bolt	BlArea	194.8383 mm2
Saddle Load Q0 (Weight)	Q0	124.4 kgf
Saddle Load QL (Wind/Seismic contribution)	QL	0.6 kgf
Maximum Transverse Force	Ft	9.0 kgf
Maximum Longitudinal Force	F1	2.3 kgf
Saddle Bolted to Steel Foundation		Yes

插管計算(Nozzle Calcs)-N1

Input, Nozzle Desc: N1

From: 20

Pressure for Reinforcement Calculations	P	8.800	kg/cm2
Temperature for Internal Pressure	Temp	100	C
Design External Pressure	Pext	1.06	kg/cm2
Temperature for External Pressure	Tempex	100	C

母材材質、應力

Parent Material		SA-403 304	
Parent Allowable Stress at Temperature	Sv	1396.80	Kg/cm2
Parent Allowable Stress At Ambient	Sva	1406.99	Kg/cm2

Inside Diameter of Cylindrical Shell	D	211.58	mm
Design Length of Section	L	64.9100	mm
Shell Finished (Minimum) Thickness	t	3.2900	mm
Shell Internal Corrosion Allowance	cas	0.0000	mm
Shell External Corrosion Allowance	co	0.0000	mm

插管計算(Nozzle Calcs)-N1

插管材質、應力

Material	SA-312 TP304		
Material UNS Number	S30400		
Material Specification/Type	Smls. & wld. pipe		
Allowable Stress at Temperature	Sn	1396.80	Kg/cm2
Allowable Stress At Ambient	Sna	1406.99	Kg/cm2
Diameter Basis (for tr calc only)	Inside		
Layout Angle		0.00	deg
Diameter		1.0000	in.
Size and Thickness Basis	Nominal		
Nominal Thickness	80S		
Flange Material	SA-182 F304		
Flange Type	Slip on		

插管計算(Nozzle Calcs)-N1

Corrosion Allowance	can	0.0000	mm
Joint Efficiency of Shell Seam at Nozzle	E1	1.00	
Joint Efficiency of Nozzle Neck	En	1.00	
插管突長、焊腳尺寸			
Outside Projection	ho	160.7450	mm
Weld leg size between Nozzle and Pad/Shell	Wo	4.0000	mm
Groove weld depth between Nozzle and Vessel	Wgnv	3.2900	mm
Inside Projection	h	0.0000	mm
Weld leg size, Inside Element to Shell	Wi	0.0000	mm
ASME Code Weld Type per UW-16		c	
Flange Class		150	
Flange Grade		GR 2.1	

插管計算(Nozzle Calcs)-N1

UG-45 Minimum Nozzle Neck Thickness Requirement: [Int. Press.]

Wall Thickness for Internal/External pressures	$t_a = 0.2134 \text{ mm}$
Wall Thickness per UG16(b),	$tr_{16b} = 1.5000 \text{ mm}$
Wall Thickness, shell/head, internal pressure	$tr_{b1} = 0.6690 \text{ mm}$
Wall Thickness	$tb_1 = \max(tr_{b1}, tr_{16b}) = 1.5000 \text{ mm}$
Wall Thickness	$tb_2 = \max(tr_{b2}, tr_{16b}) = 1.5000 \text{ mm}$
Wall Thickness per table UG-45	$tb_3 = 2.9464 \text{ mm}$

Determine Nozzle Thickness candidate [tb]:

$= \min[tb_3, \max(tb_1, tb_2)]$
 $= \min[2.946, \max(1.5, 1.5)]$
 $= 1.5000 \text{ mm}$

Minimum Wall Thickness of Nozzle Necks [tUG-45]:

$= \max(t_a, t_b)$
 $= \max(0.213, 1.5)$
 $= 1.5000 \text{ mm}$

Available Nozzle Neck Thickness = $0.875 * 4.547 = 3.978 \text{ mm}$ -> OK

UG-36(c)(3)

(3) Openings in vessels not subject to rapid fluctuations in pressure do not require reinforcement other than that inherent in the construction under the following conditions:

(-a) welded, brazed, and flued connections meeting the applicable rules and with a finished opening not larger than:

(-1) $3\frac{1}{2}$ in. (89 mm) diameter — in vessel shells or heads with a required minimum thickness of $\frac{3}{8}$ in. (10 mm) or less;

(-2) $2\frac{3}{8}$ in. (60 mm) diameter — in vessel shells or heads over a required minimum thickness of $\frac{3}{8}$ in. (10 mm);

(-b) threaded, studded, or expanded connections in which the hole cut in the shell or head is not greater than $2\frac{3}{8}$ in. (60 mm) diameter;

(-c) no two isolated unreinforced openings, in accordance with (-a) or (-b) above, shall have their centers closer to each other than the sum of their diameters;

母材厚度小於 $\frac{3}{8}$ "(10mm)，完工開孔直徑最大可達 $3\frac{1}{2}$ " (89 mm)

母材厚度大於 $\frac{3}{8}$ "(10mm)，完工開孔直徑最大可達 $2\frac{3}{8}$ " (60 mm)

任兩個「免計算的獨立開孔」，其中心距不得小於兩孔直徑的總和。

管板強度計算(ASME TS Calc: DESIGN)-胴/管側基本資料

Input Echo, Tubesheet Number 1, Description: Fixed TS

Shell Data:

Main Shell Description: Shell

Shell Maximum Design Pressure	Psd,max	6.00 kg/cm2
Shell Maximum Operating Pressure	Psox,max	3.00 kg/cm2
Shell Minimum Operating Pressure	Psox,min	-0.53 kg/cm2
Shell Thickness	ts	3.7600 mm
Shell Internal Corrosion Allowance	cas	0.0000 mm
Shell External Corrosion Allowance	casext	0.0000 mm
Inside Diameter of Shell	Ds	211.580 mm
Shell Circumferential Joint Efficiency	Esw	0.850
Shell Temperature for Internal Pressure	Ts	40.00 C
Shell Material	SA-312 TP304	

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Shell Material UNS Number	S30400
Shell Allowable Stress at Temperature	Ss 1403.40 Kg/cm2
Shell Allowable Stress at Ambient	1403.40 Kg/cm2

Note: The allowables above have been divided by 0.85, see 4.18.15 [Ss].

Channel Data:

Channel Description: Cap SF

Channel Type:

Channel Maximum Design Pressure	Ptd,max	Cylinder 8.80 kg/cm2
Channel Maximum Operating Pressure	Ptox,max	4.00 kg/cm2
Channel Minimum Operating Pressure	Ptox,min	-1.06 kg/cm2
Channel Thickness	tc	3.7600 mm
Channel Internal Corrosion Allowance	cac	0.0000 mm
Channel External Corrosion Allowance	cacext	0.0000 mm
Inside Diameter of Channel	Dc	211.580 mm
Channel Design Temperature	TEMPC	100.00 C
Channel Flange Matl	SA-403 304	

See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Channel Material UNS Number	S30400
Channel Allowable Stress at Temperature	Sc 1396.80 Kg/cm2
Channel Allowable Stress at Ambient	1406.99 Kg/cm2

管板強度計算(ASME TS Calc: DESIGN)-傳熱管數據

Tube Data:

Number of Tube Holes	Nt	36
Tube Wall Thickness	et	1.6500 mm
Tube Outside Diameter	D	19.0500 mm
Total Straight Tube Length	Lt	<u>1829.00 mm</u>
Straight Tube Length (bet. inner tubsht faces) L		1773.00 mm
Design Temperature of the Tubes		100.00 C
Tube Material	SA-213	TP304
Tube Material UNS Number		S30400
Is this a Welded Tube		No
Tube Material Specification used	Smls.	tube
Tube Allowable Stress at Temperature		1396.80 Kg/cm2
Tube Allowable Stress At Ambient		1406.99 Kg/cm2
Tube Yield Stress At design Temperature	Syt	1733.29 Kg/cm2
Tube Pitch (Center to Center Spacing)	P	25.4000 mm
Tube Layout Pattern		Triangular

管板強度計算(ASME TS Calc: DESIGN)

Fillet Weld Leg	af	1.8000 mm
Groove Weld Leg	ag	0.0000 mm
Tube-Tubesheet Joint Weld Type	Seal/No Weld	
Method for Tube-Tubesheet Jt. Allowable	UW-20.4.2	
Tube-Tubesheet Joint Classification	i	
Is Tube-Tubesheet Joint Tested	No	
ASME Tube Joint Reliability Factor	fr	0.70
Interface Pressure, after tube expansion	Po	0.00 kg/cm2
Interface Pressure, due to diff. thermal exp.	PT	0.00 kg/cm2
Radius to Outermost Tube Hole Center	ro	84.675 mm
Largest Center-to-Center Tube Distance	U1	34.0000 mm
Length of Expanded Portion of Tube	ltx	22.0000 mm
Distance from Channel Side where Exp. Starts	3.0000 mm	
Min. Distance from Shell Side where Exp. Stops	3.0000 mm	
Tube-side pass partition groove depth	hg	5.0000 mm

管板強度計算(ASME TS Calc: DESIGN)

Tubesheet Data:

Tubesheet TYPE: Fixed Tubesheet Exchanger, Conf B

Tubesheet Design Metal Temperature T 100.00 C
Tubesheet Material SA-182 F304

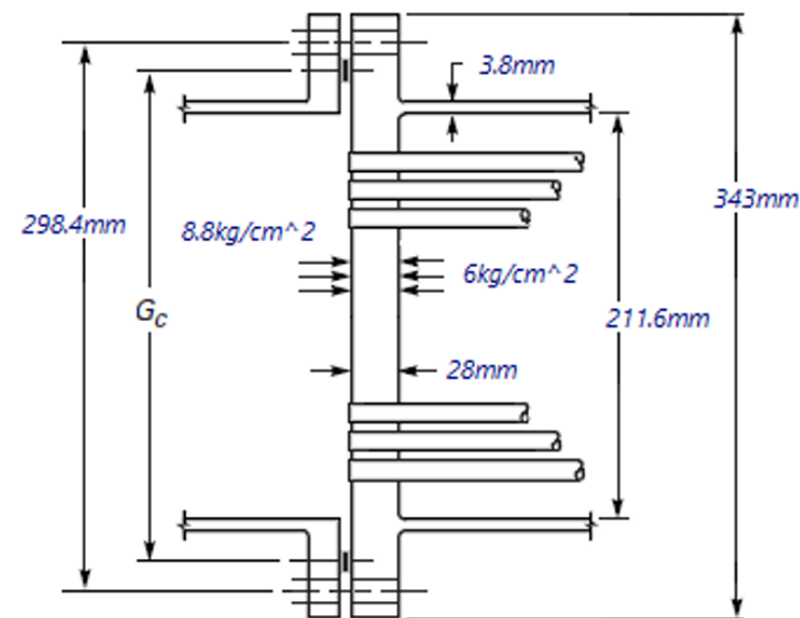
See ASME VIII Div. 1, UG-23(e), for determination of the SPS allowable.

Tubesheet Material UNS Number		S30400
Tubesheet Allowable Stress at Temperature	S	1152.14 Kg/cm ²
Tubesheet Allowable Stress at Ambient	Tt	1406.99 Kg/cm ²
Thickness of Tubesheet	h	28.0000 mm
Tubesheet Corr. Allowance (Shell side)	Cats	1.6000 mm
Tubesheet Corr. Allowance (Channel side)	Catc	1.6000 mm
Tubesheet Outside Diameter	A	343.000 mm

Additional Data for Stepped Tubesheets: 管板各階級厚度及外徑

Is the Tubesheet Stepped?		Yes
Is the Tubesheet Flat on Tubeside?		No
Step 1 Diameter on the Tubeside	dt1	209.00 mm
Step 1 Depth on the Tubeside	ht1	5.00 mm
Is the Tubesheet Flat on Shellside?		No
Step 1 Diameter on the Shellside	ds1	209.00 mm
Step 1 Depth on the Shellside	hs1	2.00 mm
Calculated Tubesheet Diameter per Div 2, 4.18.4		343.00 mm

Tubesheet Integral w/Shell, Gasketed w/Channel, Extended as a Flange



Configuration b:

管板強度計算(ASME TS Calc: DESIGN)

Additional Data for Fixed/Floating Tubesheet Exchangers:

Unsupported Tube Span under consideration	l	108.000 mm
Tube End condition corresponding to Span (l)	k	1.00
Tubesheet Metal Temp. at Rim	T'	15.00 C
Shell Metal Temp. at Tubesheet	T'S	5.00 C
Channel Metal Temp. at Tubesheet	T'C	25.00 C
Perform Differential Pressure Design		No
Run Multiple Load Cases		Yes
Shell Side Min. Design Pressure	Psd,min	-0.5300 kg/cm2
Channel Side Min. Design Pressure	Ptd,min	-1.0600 kg/cm2
Mean Shell Metal Temp. along Shell len.	Tsm	5.00 C
Mean Tube Metal Temp. along Tube length	Ttm	15.00 C
Junction Stress Reduction option	Perform Plastic Calculation	

傳熱管未被支撐的長度

管板強度計算(ASME TS Calc: DESIGN)

Additional Data for Gasketed Tubesheets:

Tubesheet Gasket on which Side		Channel	
Flange Outside Diameter	A	343.000	mm
Flange Inside Diameter	B	211.580	mm
Flange Face Outside Diameter	Fod	269.700	mm
Flange Face Inside Diameter	Fid	211.580	mm
Gasket Outside Diameter	Go	279.000	mm
Gasket Inside Diameter	Gi	219.000	mm
Small end Hub thk.	g0	3.7600	mm
Large end Hub thk.	g1	17.2100	mm
Gasket Factor,	m	2.00	
Gasket Design Seating Stress	y	112.48	Kg/cm2
Flange Facing Sketch	Code	Sketch 1a	
Gasket Thickness	tg	3.2000	mm
Full face Gasket Flange Option	Program	Selects	
Length of Partition Gasket	lp	409.000	mm
Width of Partition Gasket	wp	6.0000	mm
Partition Gasket Factor,	mPart	2.0000	
Partition Gasket Design Seating Stress	yPart	112.48	Kg/cm2
Partition Gasket Facing Sketch	Code	Sketch 1a	
Partition Gasket Column for Gasket Seating	Code	Column II	

管板墊片用在哪一側

管板強度計算(ASME TS Calc: DESIGN)

Bolting Information:

Diameter of Bolt Circle	C	298.400	mm
Nominal Bolt Diameter	dB	19.0500	mm
Thread Series	TEMA Thread Series		
Number of Bolts	n	12	
Bolt Material		SA-193 B7	
Bolt Allowable Stress At Temperature	Sb	1753.64	Kg/cm2
Bolt Allowable Stress At Ambient	Sa	1753.64	Kg/cm2
Weld between Flange and Shell/Channel		4.0000	mm
Tubesheet Integral with		Shell	
Tubesheet Extended as Flange		Yes	
Thickness of Extended Portion of Tubesheet	Tf	21.0000	mm
Is Bolt Load Transferred to the Tubesheet		Yes	
Is Exchanger in Creep range (skip EP, Use 3S for Sps)		No	

焊接填角焊尺寸

管板與胴側焊再一起

管板擴展成法蘭(外頭由螺栓孔與法蘭對鎖)

簡報結束 敬請指教

